

Loyola Marymount University

CMSI 3300 Artificial Intelligence

Fall 2024

Professor Lanyu Shang

Announcements

- Homework 2
  - Due next Friday 10/25 @5 pm
  - No other task this week and try to focus on it!
  - Will have time to work on it today

Midterm Score

- Review in-class today
- You may also come to my office hours to review your exam.
- The scores will be **curved** and available on Brightspace soon.
- Final exam is **non-cumulative**.
- Exams are also weighted.
  - Higher exam score -> 2/3 exam grade (35% of final grade)
  - Lower exam score -> 1/3 exam grade
- Still time and opportunities for improvement!

Test-Taking Strategies

- Show your work
  - Write out your thinking process (even on scratch paper).
  - Partial credit is often given for approach
  - Don’t skip multiple choice questions
- Cheat sheet
  - Organize information logically – don’t get overloaded
  - Prioritize content
  - Practice using your cheat sheet

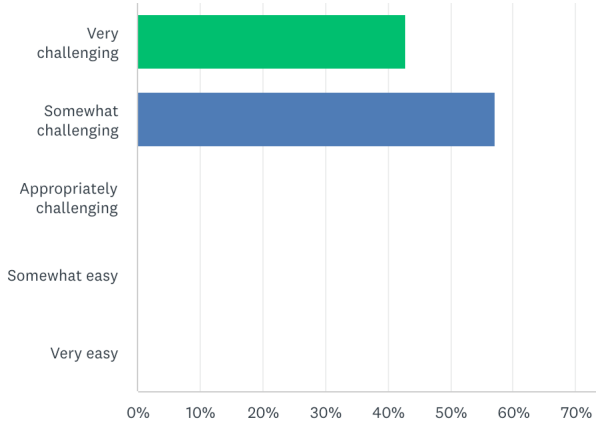
Resources

- Office Hours
  - Tuesdays/Thursdays 10 AM – Noon, Doolan 205
  - By appointment
- Recitations
  - Tuesdays @6 PM, Keck Annex
- TA office hours
  - Keck Annex



Feedback from the Survey

- The materials are challenging
  - Opportunity to practice and examine your (life-long) learning skills
- More real-world examples
  - Yes!
- Pace of the lecture
  - Can be faster and have time for Q&A
  - Can be slowed down a bit and have more explanations
  - Resolution: your participation!



The survey will remain open. Your feedback is crucial for improving the course before the semester ends.

Reminders and Updates

- Maintaining the Keck Lab
  - Please keep the lab space clean and tidy
- Group Changes
  - If you want to change your group for HW/CW, please inform me **at least a week before the deadline**.

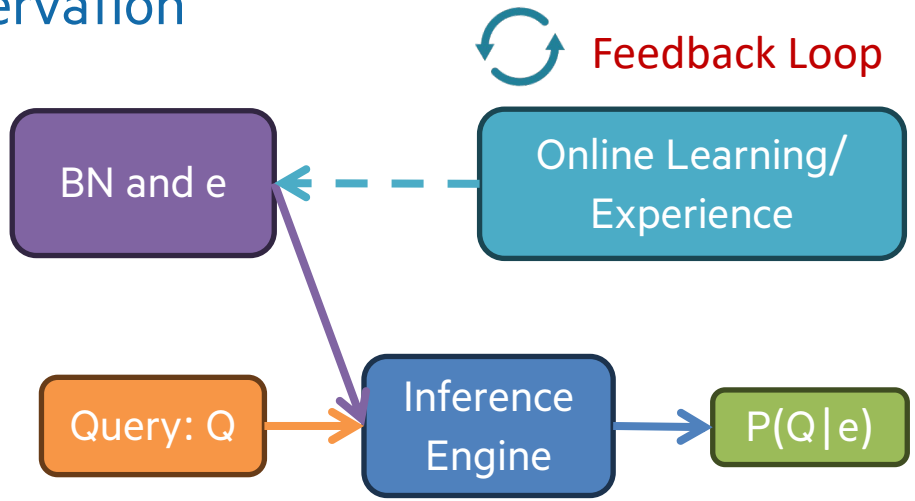


This Lecture

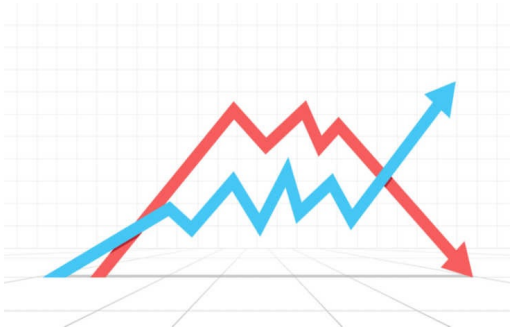
- Markov Models
  - Reasoning over time and space
- Homework 2

Bayesian Networks & Probabilistic Reasoning
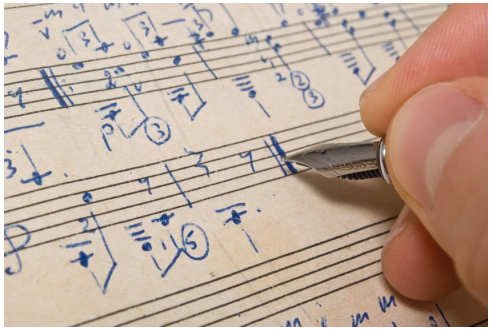
- Assumption: evidence is frozen in time
- What is the limitation?
  - New experience/observation
- Any examples?



Examples

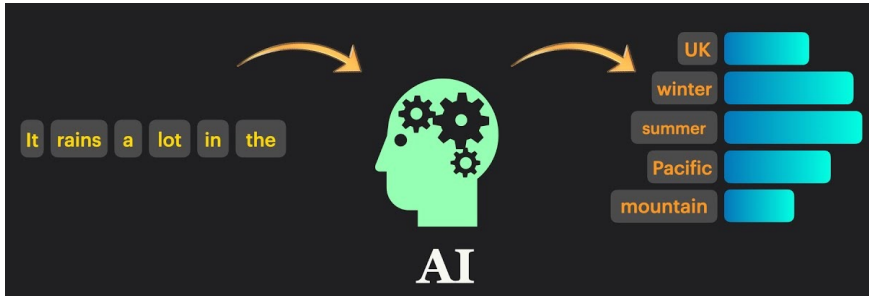


Stock Market



Music Composition





Text Generation

Google Pagerank

loganix

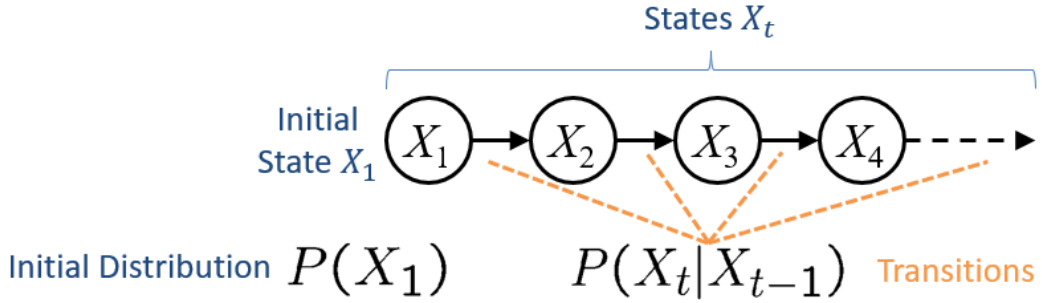


Markov Models

- Markov Models / Markov Chains:
  - Model the **changes in some State Variables** over time/space
    - X_t where X is the state variable and t is the time step.
  - Specified by **transition probabilities** (a.k.a. dynamics)
    - How state changes over time
    - Transition probabilities are assumed to be **stationary**
      - The probability moving **from one state to next** is **fixed**

Markov Models

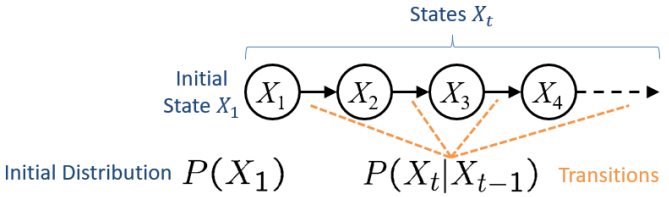
- Markov Model (MM) as a BN



- Property
  - BN: the probability of a child only depends on its parents.
  - MM: The **future** is **independent** of the past given the present.

Why?

Joint Distribution Factorization



- Factor the following joint distribution in MM

$$P(X_1, X_2, \dots, X_T)$$
$$= P(X_1)P(X_2|X_1)P(X_3|X_2) \dots P(X_T|X_{T-1})$$
$$= P(X_1) \prod_{t=2}^{T} P(X_t|X_{t-1})$$

- Like BN, we only need the transition probability to compute the joint probability.

Mini-Forward Algorithm

- How about answering arbitrary $P(X_t)$?
- Mini-Forward Algorithm

$$P(X_t = x_t) = \sum_{x_{t-1} \in X_{t-1}} P(X_t = x_t | X_{t-1} = x_{t-1}) P(X_{t-1} = x_{t-1})$$

transition probability

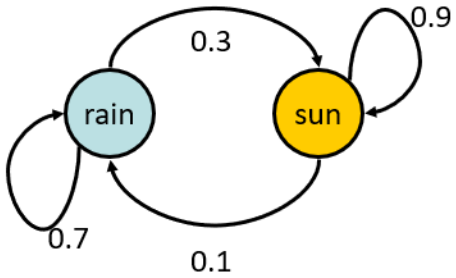
state likelihood at **t-1**

All possibilities of previous state

Example

- Suppose
  - $X = \{rain, sun\}$
  - $P(X_1 = sun) = 1$

X_{t-1}	X_t	$P(X_t \mid X_{t-1})$
sun	sun	0.9
sun	rain	0.1
rain	sun	0.3
rain	rain	0.7



- Given the MM, compute $P(X_3 = sun)$,

  i.e., the likelihood that it will be sunny in 2 days.

Example

- $P(X_3 = sun) = \sum_{x_2 \in X_2} P(X_3 = sun | X_2 = x_2) P(X_2 = x_2)$
- We will need $P(X_2 = x_2)$ for all $x_2 \in X_2$
  - $P(X_2 = sun) = \sum_{x_1 \in X_1} P(X_2 = sun | X_1 = x_1) P(X_1 = x_1)$
    $= P(X_2 = sun | X_1 = sun) P(X_1 = sun) +$
    $P(X_2 = sun | X_1 = rain) P(X_1 = rain)$
    $= 0.9 * 1.0 + 0.3 * 0 = 0.9$
  - $P(X_2 = rain) = 1 - 0.9 = 0.1$

Example

- $P(X_3 = sun) = \sum_{x_2 \in X_2} P(X_3 = sun | X_2 = x_2) P(X_2 = x_2)$

$$= P(X_3 = sun | X_2 = sun) P(X_2 = sun) +$$

$$P(X_3 = sun | X_2 = rain) P(X_2 = rain)$$

$$= 0.9 * 0.9 + 0.3 * 0.1 = 0.84$$

Next…

HOMEWORK 2